GastroOps: Corpus & Dataset Technical Dossier

Yelp Academic Dataset Curation • Vector Embedding Index • Database Schema

1. Dataset Origin & Curation Methodology

GastroOps is powered by an operational curation of the **Yelp Academic Dataset**. The raw source collection represents an 8.7 GB JSON corpus spanning over 6.9 million customer reviews across North America. To build a focused, production-grade restaurant intelligence platform, an intensive data curation pipeline extracted authentic restaurant operations feedback.

Attribute	Raw Source Dataset	Curated GastroOps Production Subset
Primary Source	Yelp Open Academic Dataset	Restaurant & Dining Operations Specific Subset
Total File Size	8.7 GB uncompressed JSON	Normalized Relational & Vector Stores
Reviewed Businesses	150,000+ commercial venues	**750 Unique Dining Establishments**
Indexed Reviews	6,990,280 general reviews	**1,000 Operational Yelp Reviews**
Geographic Spread	Broad metropolitan regions	Tucson, Indianapolis, Reno, Tampa, Nashville, Philadelphia

2. Sentiment & Star Rating Distribution

To ensure balanced evaluation across both critical failure remediation and operational excellence commendations, the curated dataset maintains a realistic multi-polarity rating curve:

Rating Tier	Star Rating	Percentage	Operational Role in Pipeline
Critical Friction Tier	1.0 - 2.0 ★	42.0%	Powers root-cause ABSA, P0-P2 SLAs, and corrective floor interventions
Mediocre / Mixed Tier	3.0 ★	23.0%	Evaluates pacing gaps, value perceptions, and consistency drift
Excellence Tier	4.0 - 5.0 ★	35.0%	Powers operational commendations, team shout-outs, and P4 sustaining SOPs

3. Relational & Vector Storage Architecture

PostgreSQL 14 Relational Schema:
• yelp_businesses: business_id (PK), name, city, state, stars, review_count, categories
• yelp_reviews: review_id (PK), business_id (FK), stars, date, text, useful, funny, cool

Qdrant Vector DB Collection:
• Collection Name: yelp_reviews_bge_m3
• Vector Dimension: **1024-dimensional dense float vectors**
• Embedding Model: BAAI/bge-m3 (Multi-aspect, multi-lingual bi-encoder)
• Distance Metric: Cosine Similarity with HNSW indexing (m=16, ef_construct=100)
• Payload Fields: review_id, business_id, restaurant_name, stars, date, city, state, text

4. Preprocessing & Sanitization Pipeline

Before indexing, review texts undergo strict normalization in backend/app/ingestion/yelp_ingest.py:
1. **Negation Preservation**: Protects critical negation expressions like *'never got sick'* or *'not cold'* to prevent false positive hazard flags.
2. **Entity & Metadata Enrichment**: Reviews are enriched with restaurant metadata prior to embedding: Restaurant: {name} | Category: {cat} | Customer Feedback: {text}.
3. **BM25 Lexical Inverted Indexing**: Stopword-filtered tokenization indexes exact operational terms for food safety, allergens, and kitchen ticket pacing.